

MASOOD AHMAD

Senior Data Engineering & IS Security Leader | Microsoft Fabric, Databricks & Azure | Data Warehousing, ETL & Decision Support

Dallas, TX 75054 | 469-200-0153 | masooda0001@gmail.com | linkedin.com/in/masood-is | US Citizen

PROFESSIONAL SUMMARY

Senior data engineering and IS security leader with 12+ years building and operating enterprise-scale data warehousing, ETL, and cloud data platforms across healthcare, finance, and SaaS, including 8+ years of team and project leadership. Strong hands-on expertise with Microsoft Fabric (Fabric Data Warehouse, semantic models, OneLake), Azure Databricks (Delta tables, Apache Spark), Azure Data Factory, Synapse, SQL Server, and Python (PySpark and notebooks) for data engineering, transformation logic, and automation. Leads day-to-day operation of production pipelines including monitoring, troubleshooting, performance tuning, and rapid incident resolution, and acts as final technical reviewer for major data model and pipeline changes. Directs IoMT and clinical device data modernization and SOC telemetry pipelines, aligning all data engineering with HIPAA, HITRUST, HITECH, NIST 800-53/66, and FDA requirements. Evaluates and guides practical adoption of AI and Large Language Models (LLMs) across data engineering workflows to power analytics, machine learning, and business intelligence.

TECHNICAL SKILLS

Cloud and Data Platforms: Microsoft Fabric (Fabric Data Warehouse, Fabric Semantic Models, OneLake, Notebooks), Azure Databricks, Azure Synapse Analytics, Azure Data Factory (ADF), ADLS Gen2, AWS (S3, Glue, Lambda, Athena, Redshift), Snowflake, Azure SQL, Azure Functions, Azure Event Hubs, Azure Key Vault, Azure Monitor, Databricks Unity Catalog

Data Engineering: Apache Spark, PySpark, Notebooks, Kafka, Delta Lake, Delta Tables, Apache Iceberg, dbt, Apache Airflow, Apache NiFi, Fivetran, SSIS, Data Warehousing, ETL and ELT Operations, Pipeline Monitoring, Troubleshooting, Performance Tuning, Incident Resolution, Medallion Architecture, Star Schema, CDC, Real-Time Streaming

Leadership: Team Leadership, Project Leadership, Technical Direction, Mentoring, Code and Data Model Review, Final Technical Reviewer, Cross-Functional Collaboration, Vendor Management, Cost Management, Onshore and Offshore Team Management

IoMT and Interoperability: IoMT and Medical Device Data, HL7, FHIR, Device Telemetry Ingestion, EMR and EHR Integration, API Integration, Hardware, Firmware and Embedded Device Data, SCCM, WSUS

Databases and Warehousing: Snowflake, Redshift, BigQuery, SQL Server, PostgreSQL, MySQL, Oracle, Enterprise Data Warehouse, Data Lakehouse, Star and Snowflake Schema

AI and Analytics: Azure OpenAI Service, Azure AI, Snowflake Cortex, GitHub Copilot, Large Language Models (LLMs), Great Expectations, Monte Carlo Observability, Tableau, Power BI

Security and Governance: RBAC, Encryption, HIPAA, HITRUST, HITECH, NIST 800-53/66, FDA Cybersecurity, ISO 14971, PCI-DSS, SOX, PHI and PII Protection, Azure Purview, Data Governance, SIEM (Splunk, Chronicle, Azure Sentinel), Entra ID

DevOps and Automation: Kubernetes (AKS, EKS), Docker, Terraform, Bicep, ARM Templates, Azure DevOps, GitHub Actions, Jenkins, CI/CD, Infrastructure as Code (IaC), Ansible, DevSecOps

Languages and Interoperability: Python, SQL, Bash, Java, Scala, FHIR, HL7, REST APIs

PROFESSIONAL EXPERIENCE

Data Security Manager

Baylor Scott & White | October 2024 to Present

- Oversee Azure cloud data engineering and IoMT data pipeline modernization, architecting and optimizing Azure-based ingestion pipelines for IoMT and clinical device telemetry using ADF, Synapse, ADLS, Microsoft Fabric, and automation frameworks.
- Built reusable API, HL7, FHIR, and device telemetry ingestion patterns, reducing manual integration work by 40% and improving data availability; partnered with BI, cloud, and biomedical teams to normalize and enrich IoMT datasets, increasing data reliability by 60%.
- Led day-to-day operation of production data pipelines including monitoring, troubleshooting, performance tuning, and rapid incident resolution, and acted as final technical reviewer for major data model and pipeline changes.
- Managed SOC telemetry pipelines (Splunk, Chronicle) and oversaw vulnerability, patch, and configuration data pipelines across 5,000+ connected medical devices using SCCM, WSUS, and custom ingestion tooling.
- Built Microsoft Fabric Data Warehouse and semantic models with OneLake and Delta tables, using Python, PySpark, and notebooks for transformation logic and automation to power executive dashboards and analytics.
- Evaluated and guided practical adoption of AI and Large Language Models (LLMs) across data engineering workflows including pipeline development, code assistance, testing, and documentation.
- Ensured all data engineering and integration processes aligned with HIPAA, HITRUST, HITECH, NIST 800-53/66, and FDA cybersecurity requirements, and defined enterprise governance standards for lineage, retention, metadata, auditability, and quality.

Director, Information Technology & Data

Crescent Regional Hospital | October 2015 to August 2024

- Led enterprise data engineering initiatives within IT operations, supervising a 16-member in-house IT team and an offshore data and IT team in India, and defined data strategy aligned with infrastructure modernization.
- Oversaw data readiness during new hospital acquisitions and integration of 50+ Urgent Care clinics nationwide, unifying on-premise and cloud data systems for interoperability and performance.
- Designed and managed scalable Azure-based data pipelines and ETL workflows on Databricks, Synapse, and Snowflake supporting clinical, operational, and security data flows, increasing data quality from 87% to 98%.
- Established data ingestion and governance frameworks for IoMT and EMR systems to support analytics and regulatory reporting, and developed dbt and PySpark ELT models on Snowflake and Delta Lake.
- Implemented data protection strategies using encryption, role-based access, and secure transmission protocols, integrating NIST and ISO cybersecurity controls to maintain HIPAA and HITRUST compliance.
- Managed service delivery and ticketing through Freshworks, partnered with vendors for technology selection and onboarding, and presented data operations and performance reports to senior leadership.

Data Analyst

ASP Cares | November 2014 to September 2015

- Designed and maintained scalable ETL and ELT pipelines and enterprise data warehouses supporting SaaS analytics products and high-volume application telemetry.
- Built financial data pipelines integrating transactional, billing, and revenue data, delivering financial reporting and analytics dashboards to support forecasting and reconciliation.
- Developed conceptual, logical, and physical data models; performed data modeling, reverse engineering, and system integration to unify data across multiple databases.
- Implemented version-controlled deployment pipelines and containerization (Docker) to improve reliability and accelerate delivery of data services.
- Optimized database and query performance across Oracle, SQL Server, and PostgreSQL, ensuring data integrity, security, and availability for enterprise applications.

KEY PROJECTS

IoMT and Clinical Device Data Modernization, Baylor Scott & White

- Modernized IoMT and clinical device data ingestion across 5,000+ connected devices using Azure Data Factory, Synapse, Microsoft Fabric, and reusable HL7/FHIR patterns, reducing manual integration by 40% and increasing data reliability by 60% while maintaining HIPAA, HITRUST, and FDA compliance.

Enterprise Data Platform and Acquisition Integration, Crescent Regional Hospital

- Built Azure and Snowflake data pipelines and governance frameworks supporting integration of 50+ Urgent Care clinics and hospital acquisitions, increasing data quality from 87% to 98% while leading a 16-member onshore and offshore team.

Financial and SaaS Analytics Data Warehouse, ASP Cares

- Delivered ETL pipelines and a data warehouse integrating transactional, billing, and revenue data for financial reporting and SaaS analytics, with Docker-based CI/CD automation improving reliability across enterprise applications.

EDUCATION

Executive MBA | Quantic School of Business and Technology | 2021

Master of Software Engineering (MSE) | COMSATS University

CERTIFICATIONS

Google Cybersecurity Professional Certificate

Oracle Certified Expert (OCE) - Oracle Real Application Clusters 11g and Grid Infrastructure Administrator

Oracle Certified Professional (OCP) - Oracle Database 12c

Oracle E-Business Suite R12 Applications Database Administrator Certified Professional (OCP Apps DBA R12)

Oracle Certified Professional (OCP) - Oracle Database 11g R2

Oracle Certified Associate (OCA) - Oracle Database 11g R2